EE1060: Differential Equations and Transform Techniques

Practice Problem Set - 09

Practice problems in topics covered till: Lecture 34

Note : The practice problem sets are designed for your personal review and are not required for submission or evaluation. These problems aim to reinforce your understanding of the material and help you prepare effectively. Some of the practice problems are sourced from the course references, and it is strongly recommended that you consult these references for additional practice problems. Due credit is also given to the TAs of the course, who have contributed to the creation of this problem set.

---

1. Find the Laplace Transform of the following signals:
   (a) $t^n u(t)$, where n is a positive integer and a is a constant.
   (b) $t^n e^{at} u(t)$, where n is a positive integer and a is a constant.
   (c) $t^n \sin(\omega_0 t) u(t)$, where n is a positive integer and ω_0 is a constant.
   (d) $t^n \cos(\omega_0 t) u(t)$, where n is a positive integer and ω_0 is a constant.
   (e) $e^{bt} \sin(\omega_0 t) u(t)$, where b and ω_0 are constants.
   (f) $\left(t^2 + 3t + 2\right) u(t)$
   (g) $\delta(t-2)e^{3t}$, where $\delta(t)$ is the Dirac delta function.

2. Determine the Laplace Transform of the following piecewise functions:
   (a) $f(t) = \begin{cases} t, & 0 \le t \le 2 \\ 2, & t > 2 \end{cases}$
   (b) $g(t) = e^{2t} u(t-1)$, where $u(t-1)$ is the unit step function shifted by 1.

3. Solve the following second-order differential equation using the Laplace Transform.

$$y''(t) + 5y'(t) + 6y(t) = 0, \quad y(0) = 2, \quad y'(0) = -1. \tag{1}$$

4. Solve the following second-order differential equation using the Laplace Transform.

$$y''(t) + 2y'(t) + y(t) = e^{3t}, \quad y(0) = 0, \quad y'(0) = 0. \tag{2}$$

5. Solve the second-order differential equation using the Laplace Transform.

$$y''(t) - 4y(t) = 0, \quad y(0) = 3, \quad y'(0) = 0. \tag{3}$$

6. Solve the third-order differential equation using the Laplace Transform.

$$y^{(3)}(t) + 3y''(t) + 2y'(t) = 0, \quad y(0) = 1, \quad y'(0) = 0, \quad y''(0) = -1. \tag{4}$$

7. Solve the fourth-order non-homogeneous differential equation using the Laplace Transform.

$$y^{(4)}(t) - 6y''(t) + 9y(t) = 3\sin(t), \quad y(0) = 0, \quad y'(0) = 0, \quad y''(0) = 0, \quad y^{(3)}(0) = 0. \tag{5}$$

8. Determine the impulse response of the following differential equation using the Laplace Transform.

$$y''(t) + 4y'(t) + 3y(t) = \delta(t), \quad y(0) = 0, \quad y'(0) = 0. \tag{6}$$

9. Find the impulse response of the following system using the Laplace Transform.

$$y^{(3)}(t) + 2y'(t) - y(t) = \delta(t), \quad y(0) = 1, \quad y'(0) = 0, \quad y^{(2)}(0) = 0. \tag{7}$$

10. Compute the inverse Laplace transform of the following expressions:
    (a) $\mathcal{L}^{-1}\left\{\frac{3}{s^2+4s+5}\right\}$.
    (b) $\mathcal{L}^{-1}\left\{\frac{2s+3}{s^2+6s+10}\right\}$.
    (c) $\mathcal{L}^{-1}\left\{\frac{5}{s^2-4s+13}\right\}$.
    (d) $\mathcal{L}^{-1}\left\{\frac{6}{s^2+9}\right\}$.
    (e) $\mathcal{L}^{-1}\left\{\frac{2}{s(s+2)}\right\}$.
    (f) $\mathcal{L}^{-1}\left\{\frac{3s+4}{(s^2+2s+2)(s+1)}\right\}$.
    (g) $\mathcal{L}^{-1}\left\{\frac{s+1}{s^2+2s+2}\right\}$.
    (h) $\mathcal{L}^{-1}\left\{\frac{1}{s^2+1} \cdot \frac{1}{s+2}\right\}$.
    (i) $\mathcal{L}^{-1}\left\{\frac{s+4}{s^2+4s+5}\right\}$.